

Testing the Neural-Field Fusion Hypothesis on Multiscale Spatiotemporal Signals

Extended abstract — 2026-06-29. Code @ 345c17b; full numbers in results/.

Background

Two temporal modalities each carry a single, slowly-varying instantaneous-frequency trajectory $f(t)$. A pair corresponds (label 1) iff the two modalities share the *same* $f(t)$. Three fusion strategies are in tension. **Late/mid fusion** (separate encoders → pooled latents → fuse) is cheap but pooling tends to discard the fine, temporally-localized component that carries the label. **Early fusion** (cross-attention on raw tokens) preserves everything but scales badly with sequence length. **Neural-field (NF) fusion** — the candidate — fits a per-modality implicit neural representation (INR) to each signal and fuses on the field *latents*; the reconstruction objective should force the latent to preserve the fine matching component, and querying fields in a shared continuous coordinate space should make fusion natural even across mismatched sampling grids.

We pre-registered four predictions: **P1** NF's accuracy advantage over late fusion *widens* as SNR drops and $f(t)$ becomes more nonstationary; **P2** NF approaches early-fusion accuracy at far lower *fusion* compute; **P3** $f(t)$ is more decodable (higher probe R^2) from the NF latent than from the pooled embedding; **P4** behaviour varies along the amortization spectrum. The harness is built to *falsify*.

Method — the hypotheses are encoded in the dataset, not just the models

The central design principle: **every claim is operationalized as a generative knob or invariant, and verified before any model is trained**. A synthetic generator draws $f(t)$ from a fixed prior and renders it through two *different* observation models — modality A as an FM chirp, modality B as amplitude modulation — added to a multiscale $1/f$ background across S octaves, sampled on *different* rates with timing jitter.

Design choice	Hypothesis it operationalizes	Verified by
Label = correspondence ; each modality's own $f(t)$ is <i>always</i> a fresh prior draw, so only the <i>coupling</i> differs	Forces genuine fusion : $I(\text{label};A)=I(\text{label};B)=0$ but $I(\text{label};A,B)>0$ — no unimodal shortcut can "win"	C1 unimodal classifiers at chance; C2 per-modality marginals identical across classes (MMD)
Low-energy matching component buried in $1/f$ background , with an explicit SNR knob	The regime where pooling <i>should</i> fail — "fine, temporally-localized component" → P1 (SNR axis)	C3 band-power + measured SNR matches config
Trajectory complexity knob (nonstationarity = $\max df/dt $)	Pooling loses more as $f(t)$ becomes less summarizable → P1 (nonstationarity axis)	C6 difficulty monotonic for the extractor, flat for oracle
Different observation models on different grids + jitter	The shared-continuous-coordinate advantage of fields over grid-bound encoders	C4 grids mismatch; jitter present
Ground-truth $f(t)$ saved per sample	Enables the probe that <i>factors</i> the hypothesis → P3 (does the representation keep $f(t)$?)	C5 trajectories present & aligned
Oracle (sees f) + classical wavelet-coherence baseline	Establishes the achievable ceiling and a non-neural floor, so failures are attributable	C7 oracle ■ classical ■ chance

All of C1–C7 passed on both the easy and hard configs *before* modelling, so any downstream win cannot be a dataset artifact. Four fusion families (late, early, and two amortized neural fields — **LAINR**, locality-aware per-modality INR, and **OmniField**, a cross-modal-crosstalk conditioned field) were trained at **matched budget** (params/steps/optimizer/data), logging FLOPs **separately for representation and fusion**. A linear/MLP **probe** regresses $f(t)$ from each frozen representation.

Results

- P1 — supported, both axes.** Sweeping SNR from the easy baseline, the four accuracy curves coincide at the extremes (all ≈ 0.91 at 12 dB; all ≈ 0.50 at -6 dB) and **separate in the middle**. The LAINR–late gap runs $-0.03 \rightarrow +0.19 \rightarrow +0.35$ (at 3 dB) $\rightarrow +0.31 \rightarrow +0.22 \rightarrow +0.06$. Late fusion hits chance by 3 dB while LAINR holds $0.71\text{--}0.85$ through $0/-3$ dB. The nonstationarity sweep shows the same ordering (LAINR $0.92\rightarrow 0.80$ vs late $0.92\rightarrow 0.69$, with late high-variance and LAINR near-zero variance).
- P3 — supported; the mechanism.** Even on *easy*, where fusion accuracies match (~ 0.91), $f(t)$ is far more decodable from NF latents (linear R^2 $0.92\text{--}0.95$) than from pooled late/early embeddings ($0.61\text{--}0.65$). Across the SNR sweep, each method's probe- R^2 *tracks* its accuracy collapse — the representation gap **predicts** the robustness gap. Late fusion fails at low SNR because its representation never held the matching component.
- P2 — supported, with a cost caveat.** LAINR matches early-fusion accuracy at **791K fusion FLOPs vs 342M** ($\sim 430\times$ cheaper fusion), but NF pays in *representation* FLOPs, so on **total** compute it is $\sim 2\times$ the others. The efficiency claim holds on the fusion axis, not total compute.
- P4 — LAINR wins.** The simpler per-modality field + agreement head beats the heavier cross-modal-crosstalk field everywhere off the easy endpoint (e.g. 0 dB: 0.797 vs 0.633). Extra cross-modal machinery reduced robustness here.
- Controls hold.** Shuffled-pairs training stays at chance ($0.49\text{--}0.52$) for all four families; combined with C1 this confirms the accuracies are genuine joint inference.

Discussion

In this regime the hypothesis is **supported, with a clear and mechanistically explained winner**: reconstruction-trained per-modality neural-field latents (LAINR) preserve the fine matching factor that pooling discards, yielding large robustness as SNR drops and $f(t)$ becomes nonstationary, at far lower *fusion* cost than cross-attention. The probe makes the causal story concrete — it is the *representation stage*, not the fusion head, that separates the methods.

Three honest boundaries. (i) The advantage is **regime-bounded**: it vanishes where the task is trivial (high SNR, all win) and where it is impossible (very low SNR, all fail at the floor, *below* the classical 0.61 anchor — deep fusion does not beat a model-aware demodulator there). (ii) It is **LAINR-specific**, not neural-fields in general — OmniField, a SOTA-style multimodal field, was mid-pack. (iii) The efficiency is **axis-specific** (fusion FLOPs, not total compute), the canonical amortization tension: a faithful field moves cost from fusion to field-fitting. External validity to real (non-synthetic) multimodal signals is untested, and the auto-decoded end of the amortization spectrum was not measured. The decisive evidence came not from any single accuracy number but from the *shape* of degradation across the difficulty knobs that the dataset was designed to expose.